

A Counterexample to Conjecture 00000000045

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Claude (Opus 5)[†]

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Abstract

Conjecture 00000000045 asserts that the prime-sum graph G_n has a Hamilton cycle for all sufficiently large n . It does not: G_n has a Hamilton cycle only when n is even, so every odd $n \geq 3$ is a counterexample and no threshold works. The refutation is formalized in Lean 4 against Mathlib.

1 Statement

For $n \geq 1$ let G_n be the graph with vertex set $\{1, \dots, n\}$ in which $i \sim j$ if and only if $i \neq j$ and $i + j$ is prime. Conjecture 00000000045 asserts that G_n contains a Hamilton cycle for all sufficiently large n .

A Hamilton cycle of G_n is a cyclic arrangement

$$v_0, v_1, \dots, v_{n-1}$$

of all n vertices — that is, $i \mapsto v_i$ is a bijection from $\mathbb{Z}/n\mathbb{Z}$ onto $\{1, \dots, n\}$ — such that $v_i \sim v_{i+1}$ for every i , indices read modulo n . We use this description throughout; it is the definition.

2 Parity obstruction

Lemma 1. *Let $n \geq 3$ and let $v_0, \dots, v_{n-1}$ be a Hamilton cycle of G_n . Then $v_i + v_{i+1}$ is odd for every i .*

Proof. Since $n \geq 3$, the positions i and $i + 1$ are distinct modulo n , so $v_i \neq v_{i+1}$. Both are at least 1, hence $v_i + v_{i+1} \geq 1 + 2 = 3$. By hypothesis $v_i + v_{i+1}$ is prime, and a prime greater than 2 is odd. $\square$

Theorem 2. *If G_n has a Hamilton cycle and $n \geq 3$, then n is even.*

Proof. Sum the edge labels around the cycle. Since $i \mapsto i + 1$ is a permutation of $\mathbb{Z}/n\mathbb{Z}$, we have $\sum_i v_{i+1} = \sum_i v_i$, and therefore

$$\sum_{i \in \mathbb{Z}/n\mathbb{Z}} (v_i + v_{i+1}) = 2 \sum_{i \in \mathbb{Z}/n\mathbb{Z}} v_i, \tag{1}$$

which is even. On the other hand the left-hand side is a sum of n terms, each odd by Lemma 1, so it is congruent to n modulo 2. Hence $n \equiv 0 \pmod{2}$. $\square$

Corollary 3. *Conjecture 00000000045 is false.*

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Proof. Given any threshold N , the integer $n = 2N + 3$ is odd and satisfies $n \geq N$ and $n \geq 3$. By Theorem 2, G_n has no Hamilton cycle. So no threshold has the asserted property. $\square$

Remark 4 (the same obstruction, phrased as bipartiteness). If i and j have the same parity and $i \neq j$, then $i + j$ is even and at least $1 + 3 = 4$, hence not prime. So every edge of G_n joins an odd vertex to an even vertex: G_n is bipartite, with sides

$$O = \{\text{odd } i \leq n\}, \quad E = \{\text{even } i \leq n\}, \quad |O| = \lceil n/2 \rceil, \quad |E| = \lfloor n/2 \rfloor.$$

A Hamilton cycle in a bipartite graph alternates between the sides, so it requires $|O| = |E|$, which fails for every odd n . Theorem 2 is this argument in a form convenient to formalize: the double count (1) replaces the appeal to alternation.

3 What survives

The obstruction is exactly parity, and it says nothing for even n . A direct search finds a Hamilton cycle for every even n in the range checked:

n	3	4	5	6	7	8	9	10	11
Hamilton cycle	no	yes	no	yes	no	yes	no	yes	no
n	12	13	14	15	16	17	18	19	20
Hamilton cycle	yes	no	yes	no	yes	no	yes	no	yes

So the natural repair is to restrict the conjecture to even n :

For every sufficiently large even n , G_n has a Hamilton cycle.

We do not address that statement, which is a well-studied question and, to our knowledge, open; it is closely tied to the existence of “prime circles” and is supported by extensive computation. The table above is computational evidence only and is not part of the Lean development.

4 Formal verification

The accompanying Lean 4 project formalizes the refutation against Mathlib. The conjecture is stated in the formalization, so the final theorem is a formalization of its negation rather than of facts that merely entail it.

- `cycSucc` — the cyclic successor on positions `Fin n`, with `cycSucc_injective`, `cycSucc_bijective` and `cycSucc_ne` (a position is never its own successor once $n \geq 3$).
- `HasPrimeHamiltonCycle` — a Hamilton cycle of G_n , as the cyclic arrangement described above: a permutation σ of the positions, the vertex at position i being $\sigma(i) + 1$, with $\text{Nat.Prime}((\sigma i + 1) + (\sigma(\text{cycSucc } i) + 1))$ for every i .
- `ConjectureHolds` — the conjecture.
- `even_of_hasPrimeHamiltonCycle` — Theorem 2. The reindexing (1) is Mathlib’s `Fintype.sum_bijective` applied to `cycSucc`.
- `conjecture_00000000045_false` — $\neg \text{ConjectureHolds}$, by instantiating at $n = 2N + 3$.

Primality is Mathlib’s `Nat.Prime` throughout. The project builds with `lake build` on `leanprover/lean4:v4.33.1` with Mathlib `v4.33.1`. It contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms of its own: `#print axioms conjecture_00000000045_false` reports exactly the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.